PokeDex — Architecture

A vanilla-JS multi-page Pokédex. No build step, no framework, no backend. Four static HTML pages served as-is, each booting its own script. Data comes from the public PokéAPI; the only persisted state is a list of favorite IDs in `localStorage`.

Stack

- HTML5 (semantic structure, no templating)
- CSS3 (custom properties + Flexbox/Grid + media queries)
- Vanilla JavaScript (ES6+, `fetch`, `async/await`)
- PokéAPI — `https://pokeapi.co/api/v2/pokemon` (free, no key)
- `localStorage` for favorites persistence

File map

File	Role
`index.html`	Home grid — entry point
`pokemon.html`	Detail page for a single Pokémon
`favorites.html`	Grid of saved Pokémon
`about.html`	Static info page (no JS)
`styles.css`	Shared design system (loaded by every page)
`pokemon.css`	Detail-page-only styles (hero card, stat bars, cry button)
`script.js`	Logic for `index.html`
`pokemon.js`	Logic for `pokemon.html`
`favorites.js`	Logic for `favorites.html`

Architectural pattern

Each HTML page is an **isolated mini-app**. There is no module system, no shared JS bundle — every page boots its own script from scratch. Pages "communicate" only through:

1. **Navigation** — clicking a card uses `window.location.href = "pokemon.html?name=..."` rather than client-side routing.
2. **`localStorage["pokemon-favorites"]`** — a JSON array of Pokémon IDs, read/written by all three scripts.

This is why `getFavorites()` and `saveFavorites()` are duplicated across `script.js`, `pokemon.js`, and `favorites.js`.

Page-by-page summary

`index.html` + `script.js` — Home

Loads all 1025 Pokémon names up front, then fetches details in **batches of 40** as the user scrolls. Three flags drive the state machine:

- `isLoading` — guards against duplicate fetches
- `currentOffset` — pagination cursor
- `filteredMode` — when true, search/filter is active and infinite scroll is paused

Search and type filter only operate on **already-loaded** Pokémon — they don't refetch.

`pokemon.html` + `pokemon.css` + `pokemon.js` — Detail

Reads `?name=` from the URL (defaults to `charizard`). Single API call to `/pokemon/{name}`. Render highlights:

- Gradient header built from primary + secondary type colors
- Sprite swap on hover: official artwork ↔ animated Showdown sprite
- Stat bars animated via `transition: width` and a deferred `setTimeout`
- Cry sound playback if `pokemon.cries.latest` exists

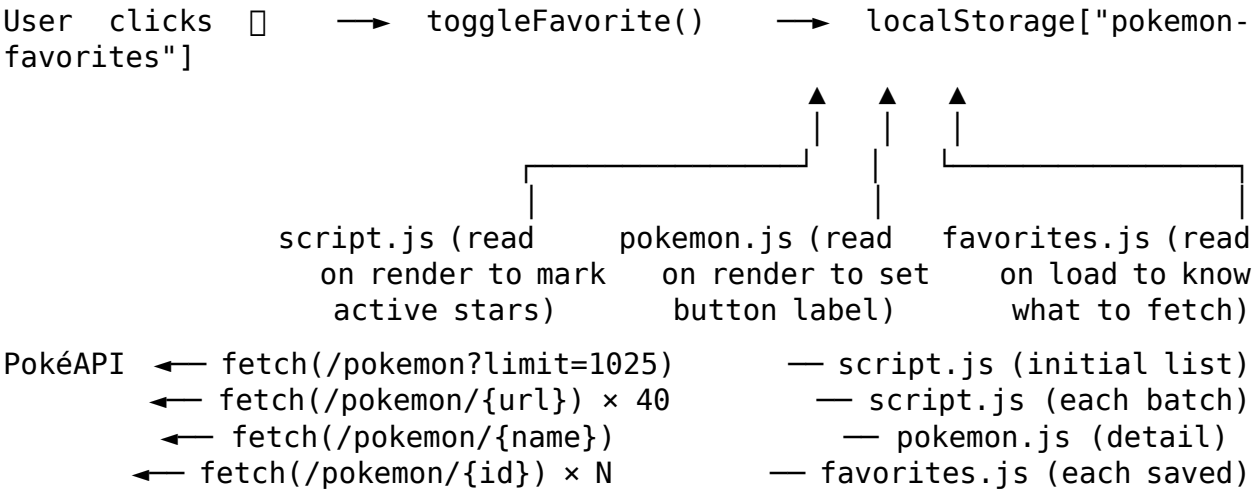
`favorites.html` + `favorites.js` — Saved

Reads IDs from `localStorage`, fetches each in parallel via `Promise.all`, renders the same card markup as the home grid. Removing a favorite mutates the DOM directly (no re-render).

`about.html` — Static

No JS. Three flexbox cards describing the stack, API, and features.

Data flow



Shared design system (`styles.css`)

CSS variables at `:root` define the theme:

```
--bg --surface --card --accent --accent2
--text --text-muted --border --shadow --radius
```

Layout is `display: grid` with `repeat(5, 1fr)` on desktop, dropping to 3 columns at ≤900px and 2 columns at ≤600px. Type badges use hardcoded per-type background colors (`.type-fire`, `.type-water`, ...).

Note: the type color palette exists in two places — as CSS classes in `styles.css` (used by grid cards) and as the `typeColors` object in `pokemon.js` (used to compute the gradient header on the detail page). Hex values differ slightly between the two.

Known quirks

- **Search blind spot** — `filterPokemon()` on the home page only searches loaded Pokémon. Searching for #500 before scrolling that far returns nothing.
- **Duplicate favorites helpers** — `getFavorites`/`saveFavorites` redeclared in three files. Candidate for extraction.
- **Stale about copy** — `about.html` says "150 Pokémon loaded" but `script.js` has `TOTAL = 1025`.
- **Orphan modal** — `index.html` and `favorites.html` include `<div id="modal">` markup, but no code path opens it. Dead leftover.
- **Sprite hover listener leak** — `pokemon.js` adds `mouseenter`/`mouseleave` listeners inside `renderPokemon()`. Searching a second Pokémon on the same page leaves the old listeners attached.
- **Inconsistent nav** — the four pages don't agree on which links to show. `pokemon.html` drops the "Search" link; `about.html` is missing it too.